
Concept Questions: Chapter 6

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ME-3640 – Heat Transfer

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Abstract

Fundamental principals and concepts are far more important to engineering expertise than understanding of example systems. These concept questions (shamelessly lifted from the textbook) will help you test your knowledge after reading the chapter summary and building a concept map. Use the relative ease and difficulty you experience while answering these questions to prioritize your efforts for the remainder of this chapter.

1 Mechanism and Types of Convection

- 6-1C** Define incompressible flow and incompressible fluid. Must the flow of a compressible fluid necessarily be treated as compressible?
- 6-2C** What is forced convection? How does it differ from natural convection? Is convection caused by winds forced or natural convection?
- 6-3C** What is external forced convection? How does it differ from internal forced convection? Can a heat transfer system involve both internal and external convection at the same time? Give an example.
- 6-4C** In which mode of heat transfer is the convection heat transfer coefficient usually higher, natural convection or forced convection? Why?
- 6-5C** Consider a hot baked potato. Will the potato cool faster or slower when we blow the warm air coming from our lungs on it instead of letting it cool naturally in the cooler air in the room? Explain.
- 6-6C** What is the physical significance of the Nusselt number? How is it defined?
- 6-7C** When is heat transfer through a fluid conduction and when is it convection? For what case is the rate of heat transfer higher? How does the convection heat transfer coefficient differ from the thermal conductivity of a fluid?

2 Boundary Layers and Flow Regimes

- 6-24C** What is the no-slip condition? What causes it?
- 6-25C** What is Newtonian fluid? Is water a Newtonian fluid?
- 6-26C** What is viscosity? What causes viscosity in liquids and in gases? Is dynamic viscosity typically higher for a liquid or for a gas?
- 6-27C** Consider two identical small glass balls dropped into two identical containers, one filled with water and the other with oil. Which ball will reach the bottom of the container first? Why?
- 6-28C** How does the dynamic viscosity of (a) liquids and (b) gases vary with temperature?
- 6-29C** What fluid property is responsible for the development of the velocity boundary layer? For what kind of fluids will there be no velocity boundary layer on a flat plate?
- 6-30C** What is the physical significance of the Prandtl number? Does the value of the Prandtl number depend on the type of flow or the flow geometry? Does the Prandtl number of air change with pressure? Does it change with temperature?
- 6-31C** Will a thermal boundary layer develop in flow over a surface even if both the fluid and the surface are at the same temperature?

- 6-32C** What is the physical significance of the Reynolds number? How is it defined for external flow over a plate of length L ?
- 6-33C** How does turbulent flow differ from laminar flow? For which flow is the heat transfer coefficient higher?
- 6-34C** What does the friction coefficient represent in flow over a flat plate? How is it related to the drag force acting on the plate?
- 6-35C** What is the physical mechanism that causes the friction factor to be higher in turbulent flow?
- 6-36C** What is turbulent viscosity? What is it caused by?
- 6-37C** What is turbulent thermal conductivity? What is it caused by?

3 Convection Equations and Similarity Solutions

- 6-47C** Consider steady, laminar, two-dimensional, incompressible flow with constant properties and a Prandtl number of unity. For a given geometry, is it correct to say that both the average friction and heat transfer coefficients depend on the Reynolds number only?
- 6-48C** Express continuity equation for steady two-dimensional flow with constant properties, and explain what each term represents.
- 6-49C** Is the acceleration of a fluid particle necessarily zero in steady flow? Explain.
- 6-50C** For steady two-dimensional flow, what are the boundary layer approximations?
- 6-51C** For what types of fluids and flows is the viscous dissipation term in the energy equation likely to be significant?
- 6-52C** For steady two-dimensional flow over an isothermal flat plate in the x -direction, express the boundary conditions for the velocity components u and v , and the temperature T at the plate surface and at the edge of the boundary layer.
- 6-53C** What is a similarity variable, and what is it used for? For what kinds of functions can we expect a similarity solution for a set of partial differential equations to exist?
- 6-54C** Consider steady, laminar, two-dimensional flow over an isothermal plate. Does the thickness of the velocity boundary layer increase or decrease with (a) distance from the leading edge, (b) free-stream velocity, and (c) kinematic viscosity?
- 6-55C** Consider steady, laminar, two-dimensional flow over an isothermal plate. Does the wall shear stress increase, decrease, or remain constant with distance from the leading edge?
- 6-56C** What are the advantages of nondimensionalizing the convection equations?
- 6-57C** Under what conditions can a curved surface be treated as a flat plate in fluid flow and convection analysis?

4 Momentum and Heat Transfer Analogies

6–82C How is Reynolds analogy expressed? What is the value of it? What are its limitations?

6–83C How is the modified Reynolds analogy expressed? What is the value of it? What are its limitations?

A Rubrics

Submissions of this assignment will be graded upon the criteria found in this section. Review these carefully, and complete your work so that application of this rubric produces your desired score.

 Criterion 1 Problems	
<i>Homework is completed with appropriate effort, neatness, etc.</i>	
Points	Description
20	Honest effort to participate and learn: every problem shows an answer or questions which need resolution.
10	Rushed or haphazard work: some problems show no effort at all.
0	Work is absent, wholly incorrect, or plagiarized.